Course Code: CS3242 **Name of the Programme: B. Tech (H)** **Name of the Course: MLOps with Cloud Computing** **Semester:6**		
Course credit	**No. of hours per week (L + T + P)**	**Total no. of Teaching hours**
3	3(2:0:2)	60
Course Objectives:		
a) To provide foundational understanding of MLOps principles and their role in scalable ML systems.		
b) Apply cloud computing models and Infrastructure as Code (IaC) for deploying scalable ML application		
c) Automate ML pipelines using CI/CD tools and frameworks, including deployment on cloud platforms.		
d) Utilize AutoML tools and cloud-native AI APIs for efficient model development and integration.		

Course outline (Syllabus of the course) - Template - 1

Syllabus:

Module – 1 Foundations of MLOps	**No. of hours** **6**
Machine Learning Engineering Overview & Architecture, Microservices in ML: Monolithic vs Microservice Design, Introduction to MLOps Concepts Engineering Project Lifecycle, MLOps vs DevOps vs DataOp, Introduction to ML Microservices and API development, Cloud computing models (IaaS, PaaS, SaaS) for ML workloads, Setting up Cloud Developer Workspaces (GitHub Codespaces, GPU Templates), Infrastructure as Code (IaC): Terraform	

Module – 2 Continuous Delivery & Deployment in Cloud Environments	**No. of hours** **6**
Continuous Delivery for ML Applications, Data Drift, Automating ML deployment with Flask + AWS App Runner (PaaS), Introduction to CI/CD with ML pipelines, Kubeflow Architecture, Kubeflow Distributions, MLOps Pipelines in the Cloud Edge ML Overview and Use Cases (AWS, GCP), Data and model versioning with DVC and MLflow, Hardware inference and vision-based edge ML	

Module – 3 Automated Machine Learning (AutoML) Platforms and Applications	**No. of hours 6**
AutoML, Overview and Benefits, No-Code/Low-Code Platforms: Create ML, Azure ML Studio, Ludwig AutoML: Deep Dive and Use Cases, AutoML for Computer Vision and NLP (Apple, GCP, Azure), Comparative study of AutoML across cloud provider	

Module – 4 Cloud-Based AI APIs and Cloud-Native MLOps Tools	**No. of hours 6**
Introduction to AI APIs (AWS, GCP, Azure), Using AWS Comprehend (NLP), Recognition (CV), GCP AutoML APIs: Vision & NLP, Azure AutoML APIs for Prediction and CV, Core Components of Cloud Applications, Developing and Testing API Endpoints for ML, Cloud-Native MLOps Tools: MLflow, Vertex AI, AzureML, Model Registry	

Module – 5 Real-World MLOps, Ethics, and Capstone Project	**No. of hours 6**
MLOps in Production: Governance, Logging, Monitoring, MLOps Maturity Models and Trends, ML Model Serving & Data Infrastructure, Infrastructure Fairness, Data Privacy, and Bias Mitigation, Capstone, Project Design: Build, Deploy, Monitor, Scale.	

Lab Programs

Sl.No.	Name of Program/ Experiment
1	Setup and Explore Cloud Developer Workspaces
2	Create and Deploy a Simple ML Microservice API
3	Implement Infrastructure as Code (IaC) using Terraform
4	Build a CI/CD Pipeline for ML Model Deployment
5	Automate Model Versioning and Data Versioning with DVC and MLflow
6	Deploy ML Model using AWS App Runner with Flask Application
7	Create AutoML Workflow Using Cloud
8	Consume Cloud AI APIs for NLP and Computer Vision
9	Build and Deploy Kubeflow Pipelines on Kubernetes
10	Capstone Mini-Project: End-to-End MLOps Pipeline

Course outcomes
a) Demonstrate a clear understanding of MLOps concepts and the ML engineering lifecycle.
b) Apply cloud-based CI/CD pipelines and Infrastructure as Code tools to develop and deploy machine learning models effectively.
c) Implement automated machine learning workflows using AutoML platforms across different cloud providers
d) Illustrate and utilize cloud-native AI APIs and tools for building scalable ML applications
e) Design and execute end-to-end MLOps projects with a focus on monitoring, governance, and ethical considerations

Reference books	
1	Kastner, Christian. Machine learning in production: from models to products. MIT Press,2025, ISBN: 9780262049726
2	Gift, Noah, and Yaron Haviv. Implementing MLOps in the Enterprise: A Production-First Approach. O'Reilly Media, 2023. ISBN-13: 978-1098136581
3	Ameisen, Emmanuel. Building Machine Learning Powered Applications. O'Reilly Media, 2020. ISBN-13: 978-1492045111
4	Grant, Trevor, et al. Kubeflow for Machine Learning: From Lab to Production. O'Reilly Media, 2020. ISBN-13: 978-1492050115
5	https://www.deeplearning.ai/courses/machine-learning-in-production/